

SECTION A: MULTIPLE CHOICE QUESTIONS - 10 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. START WITH SECTION A AND HAND THE OPTIC READER FORM IN AFTER 50 MINUTES.
2. First circle your answers on this paper and only mark the answers on the form when you are certain of your choice. You may not erase on the optic reader form.
3. Do all scribbling on the facing page. It will not be marked.
4. Use a soft pencil to write on the optic reader form.
5. Use **SIDE 2** of the optic reader form.
Fill in your personal information in pencil on side 2.
Fill in your student number from top to bottom and then code it.
6. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 10 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. BEGIN MET AFDELING A EN HANDIG DIE MERKLEESVORM IN NA 50 MINUTE.
2. Omkring jou antwoorde eers in hierdie vraestel en merk die antwoorde op die merkleesvorm eers as jy seker is van jou keuse. Jy mag nie uitvee op die merkleesvorm nie.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik 'n sagte potlood om op die merkleesvorm te skryf.
5. Gebruik **KANT 2** van die merkleesvorm.
Vul u persoonlike inligting in potlood in op kant 2.
Vul u studentnommer van bo na onder in en kodeer dit.
6. U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

If it is given that the function $f(x) = xe^{ax}$ has a critical number at $x = 3$, then $a =$

As dit gegee is dat die funksie $f(x) = xe^{ax}$ 'n kritieke getal by $x = 3$ het, dan is $a =$

(a)	3	(b)	$\frac{1}{3}$	(c)	$-\frac{1}{3}$
(d)	$-\frac{1}{2}$	(e)	a does not exist / a bestaan nie	(f)	None of these / Geen van hierdie

Question 2 / Vraag 2

If $S(x) = \ln x$, $g(3) = 4$ and $g'(3) = 5$, determine $(S \circ g)'(3)$.

Indien $S(x) = \ln x$, $g(3) = 4$ en $g'(3) = 5$, bepaal $(S \circ g)'(3)$.

(a)	$\frac{5}{4}$	(b)	$\frac{5}{3}$	(c)	$4 \ln 3$
(d)	$5 \ln 3$	(e)	$\frac{20}{3}$	(f)	None of these / Geen van hierdie

Question 3 / Vraag 3

Choose a statement that completes the following definition: A function f is decreasing on an interval $I = [a, b]$ if

Kies 'n bewering om die volgende definisie te voltooi: 'n Funksie f daal op 'n interval $I = [a, b]$ indien

(a)	$f' < 0$ on the interval I
	$f' < 0$ op die interval I
(b)	$f(x_2) < f(x_1)$ for any $x_1, x_2 \in I$ where $x_2 < x_1$
	$f(x_2) < f(x_1)$ vir enige $x_1, x_2 \in I$ waar $x_2 < x_1$
☒ (c)	$f(x_2) < f(x_1)$ for any $x_1, x_2 \in I$ where $x_2 > x_1$
	$f(x_2) < f(x_1)$ vir enige $x_1, x_2 \in I$ waar $x_2 > x_1$
(d)	$f(b) < f(a)$
(e)	None of these / Geen van hierdie

Question 4 / Vraag 4

Given / Gegee: $f(x) = \begin{cases} x + 2 & \text{if / as } x \leq 3 \\ 4x - 7 & \text{if / as } x > 3 \end{cases}$

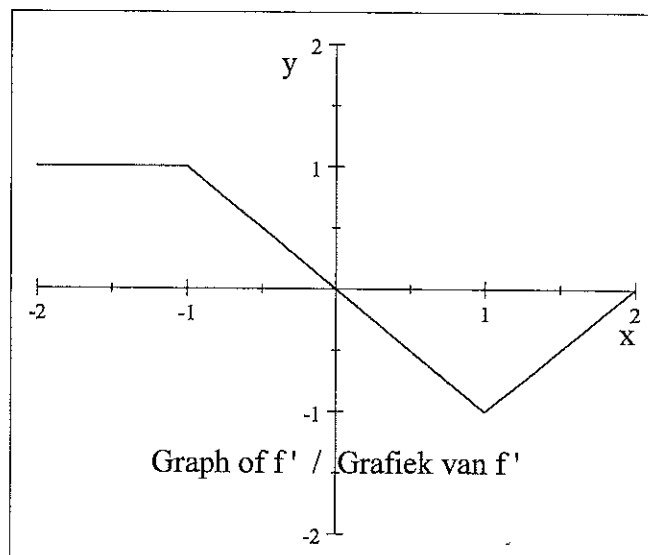
Which of the following statements is true? / Watter van die volgende stellings is waar?

- I $\lim_{x \rightarrow 3} f(x)$ exist. / $\lim_{x \rightarrow 3} f(x)$ bestaan.
- II f is continuous at $x = 3$ / f is kontinu by $x = 3$
- III f is differentiable at $x = 3$ / f is differensieerbaar by $x = 3$

(a)	I only / Slegs I	(b)	II only / Slegs II
(c)	III only / Slegs III	☒ (d)	I and II only / Slegs I en II
(e)	I and III only / Slegs I en III	(f)	I, II and III / I, II en III

Question 5 / Vraag 5

Given the graph of f' , the derivative of a function f . / Gegee die grafiek van f' , die afgeleide van 'n funksie f .



Which of the following statements is true about f ? / Watter van die volgende stellings is waar vir f ?

(a)	f is not differentiable at $x = -1$ and $x = 1$
	f is nie differensieerbaar by $x = -1$ en $x = 1$ nie
(b)	f is decreasing on $-1 \leq x \leq 1$
	f is dalend op $-1 \leq x \leq 1$
(c)	f is increasing on $1 \leq x \leq 2$
	f is stygend op $1 \leq x \leq 2$
☒ (d)	f has a local maximum at $x = 0$
	f het 'n lokale maksimum by $x = 0$
(e)	f has a local minimum at $x = 0$
	f het 'n lokale minimum by $x = 0$

Question 6 / Vraag 6

Let f be a function with derivative function $f'(x) = x - \frac{4}{x}$. On which of the following intervals is f increasing?

Laat f 'n funksie wees met afgeleide funksie $f'(x) = x - \frac{4}{x}$. Op watter van die volgende intervalle sal f stygend wees?

(a)	$(-\infty, 0) \cup (0, -\infty)$	(b)	$(-\infty, 0) \cup (0, 2)$	☒ (c)	$(-2, 0) \cup (2, \infty)$
(d)	$(-2, 2) \cup (2, \infty)$	(e)	$(-\infty, -2) \cup (0, 2)$	(f)	None of these / Geen van hierdie

Question 7 / Vraag 7

The horizontal line $y = 3$ is an asymptote for the graph of the function f for $x \geq 0$. Which of the following statements must be true?

Die horisontale lyn $y = 3$ is 'n asimptoot vir die funksie f vir $x \geq 0$. Watter van die volgende stellings moet waar wees?

(a) $f(0) = 3$	(b) $f(x) \neq 3$ for all / vir alle x	(c) $f(3)$ is undefined / is ongedefinieerd
(d) $\lim_{x \rightarrow 3} f(x) = \infty$	☒ (e) $\lim_{x \rightarrow \infty} f(x) = 3$	(f) None of these / Geen van hierdie

Question 8 / Vraag 8

The Mean Value Theorem does not apply to $f(x) = |x - 3|$ on $[1, 4]$ because:

Die Middelwaardestelling geld nie vir $f(x) = |x - 3|$ op $[1, 4]$ omdat:

(a) f is not continuous on $[1, 4]$ / f is nie kontinu op $[1, 4]$ nie
☒ (b) f is not differentiable on $(1, 4)$ / f is nie differensieerbaar op $(1, 4)$ nie
(c) $f(1) \neq f(4)$
(d) $f(1) > f(4)$
(e) $f(1) < f(4)$
(f) None of these / Geen van hierdie

Question 9 / Vraag 9

Given / Gegee: $f(3) = 3$, $f'(3) = 2$, $g(3) = -4$ and / en $g'(3) = 3$.

$$\left(\frac{f}{g}\right)'(3) =$$

(a) $\frac{1}{16}$	(b) $\frac{1}{9}$	(c) $-\frac{17}{9}$
☒ (d) $-\frac{17}{16}$	(e) $\frac{17}{16}$	(f) None of these / Geen van hierdie

Question 10 / Vraag 10

$$\lim_{t \rightarrow \infty} \frac{t - t\sqrt{t}}{2t^{3/2} + 3t - 5} =$$

☒ (a) $-\frac{1}{2}$	(b) $-\frac{1}{3}$	(c) 0
(d) $-\frac{1}{5}$	(e) $-\frac{1}{6}$	(f) None of these / Geen van hierdie

SECTION B: 40 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer the following questions on this paper **in ink**.
2. Work done in pencil or in red ink will not be marked.
3. Do not use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
If you need more space for an answer, use the facing page and indicate it clearly **by framing it**.
5. Show all steps and calculations and explain your answers.

AFDELING B: 40 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel **in ink**.
2. Werk wat in potlood of rooi ink beantwoord is, word nie nagesien nie.
3. Korrigeer-vloeistof ("Tipp-Ex") mag nie gebruik word nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
As jy nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek**.
5. Toon alle stappe en berekeninge en verduidelik jou antwoorde.

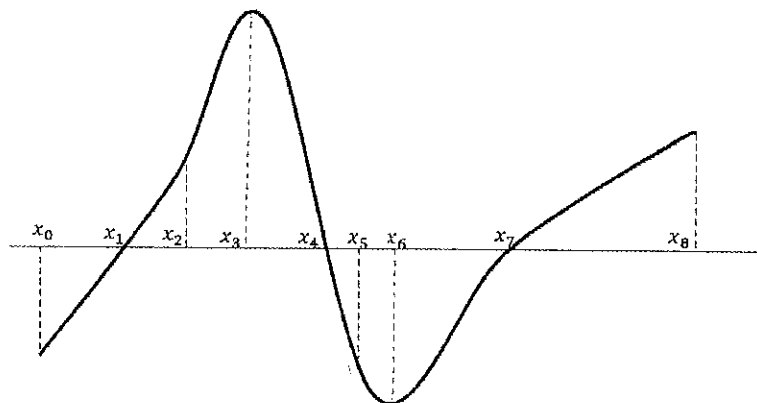
Question 1 / Vraag 1

The information below applies to Question 1.

Die onderstaande inligting is van toepassing op Vraag 1.

The graph of the derivative f' of a function f is given below.

Die grafiek van die afgeleide f' van 'n funksie f is hieronder gegee.



For 1.1 to 1.3 **write down the answer only** using interval notation where necessary.
Find all the values of x at which:

Vir 1.1 tot 1.3 **skryf slegs die antwoord neer** en gebruik intervalnotasie waar nodig.
Vind al die waarde(s) van x waar:

1.1 f is increasing / f stygend is

$$x \in (x_1, x_4) \cup (x_6, x_8)$$

1.2 f has critical points / f kritieke punte het

x_1, x_4, x_7

1.3 f has a local minimum / f 'n lokale minimum het

x_1, x_7

3

Question 2 / Vraag 2

Fill in the answer only in the space provided. Rough work can be done on the opposing page and will not be marked.

Vul slegs die antwoord in die toegekende spasie. Berekeninge kan op die teenblad gedoen word en sal nie nagesien word nie.

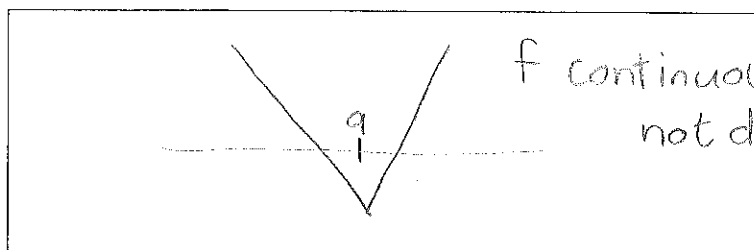
2.1 The function f is not differentiable at $x = a$. Does this imply that f is discontinuous at $x = a$?
Use a graph to illustrate your answer.

Die funksie f is nie differensieerbaar by $x = a$ nie. Impliseer dit dat f diskontinu is by $x = a$?
Gebruik 'n grafiek om jou antwoord te illustreer.

Yes / Ja

☐

No / Nee

☒


f continuous but
not diff. at $x = a$

1

2.2 f is a differentiable function with $f(3) = 4$ and $f'(3) = -3$ and $g(x) = x^2 f(x)$.

The equation of the tangent line to the graph of g at the point where $x = 3$ is given by:

f is 'n differensieerbare funksie met $f(3) = 4$ en $f'(3) = -3$ en $g(x) = x^2 f(x)$.

Die vergelyking van die raaklyn aan g by $x = 3$ word gegee deur:

$$y = -3x + 45$$

OR

$$y - 36 = -3(x - 3)$$

2

2.3 If / As $f(x) = \ln(x + 5 + e^{-5x})$, then / dan is $f'(0) =$

$$-\frac{2}{3}$$

1

2.4 The critical number(s) of $f(x) = 3x^{5/3} - 15x^{2/3}$ is:

Die kritieke getal(le) van $f(x) = 3x^{5/3} - 15x^{2/3}$ is:

$$x = 0, x = 2$$

2.5 If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{2\pi}{3}$, with $|\vec{a}| = 6$ and $|\vec{b}| = 5$, then $\vec{a} \cdot \vec{b} =$

As $\frac{2\pi}{3}$ die hoek tussen $\vec{a}$ en $\vec{b}$ is, met $|\vec{a}| = 6$ en $|\vec{b}| = 5$, dan is $\vec{a} \cdot \vec{b} =$

$$-15$$

Question 3 / Vraag 3

Determine the following derivatives. Do not simplify your answers.

Bepaal die volgende afgeleides. Moenie jou antwoorde vereenvoudig nie.

3.1 $f(x) = \tan^{-1}\left(\frac{1}{x}\right)$ (Notation / Notasie: $\tan^{-1} = \arctan = \text{bgtan}$)

$$f'(x) = \frac{-\frac{1}{x^2}}{1 + \left(\frac{1}{x}\right)^2}$$

3.2 $f(t) = \cos(\cosh t)$

$$f'(t) = -\sin(\cosh t) \cdot \sinh t$$

3.3 $f(x) = \ln\left(\frac{1}{\sqrt{x+\sqrt{x}}}\right)$ (Use the properties of $\ln$ / Gebruik die eienskappe van $\ln$).

$$= \ln 1 - \frac{1}{2} \ln(x + \sqrt{x})$$

$$f'(x) = -\frac{1}{2} \left(\frac{1 + \frac{1}{2\sqrt{x}}}{x + \sqrt{x}} \right)$$

3.4 $f(x) = (1+x)^{1/x}$

$$\ln f(x) = \frac{1}{x} \ln(1+x)$$

$$\frac{f'(x)}{f(x)} = -\frac{1}{x^2} \ln(1+x) + \frac{1}{x} \frac{1}{1+x}$$

$$f'(x) = (1+x)^{1/x} \left[-\frac{1}{x^2} \ln(1+x) + \frac{1}{x(1+x)} \right]$$

3

3.5 If $\sin(x+2y) = e^y$, find $\frac{dy}{dx}$ in terms of x and y .

Indien $\sin(x+2y) = e^y$, vind $\frac{dy}{dx}$ in terme van x en y .

$$\cos(x+2y)[1+2y'] = e^y \cdot y'$$

$$y'[e^y - 2\cos(x+2y)] = \cos(x+2y)$$

$$y' = \frac{\cos(x+2y)}{e^y - 2\cos(x+2y)}$$

3

Question 4 / Vraag 4

Let f be the function defined by / Laat f die funksie wees gedefinieer deur

$$f(x) = \begin{cases} x+2 & \text{if / as } x \leq a. \\ x^2 & \text{if / as } x > a. \end{cases}$$

Find the value(s) of a such that the function f is continuous at $x = a$.

Vind die waarde(s) van a sodat die funksie f kontinu is by $x = a$.

f continuous
at $x=a$

$$\Rightarrow f(a) = \lim_{x \rightarrow a^-} (x+2) = \lim_{x \rightarrow a^+} x^2$$

$$\Rightarrow a+2 = a^2$$

$$\Rightarrow a^2 - a - 2 = 0$$

$$\Rightarrow (a+1)(a-2) = 0$$

$$\Rightarrow a = -1 \text{ or } a = 2$$

$$\Rightarrow f \text{ continuous if } a = -1 \text{ or } a = 2.$$

3

Question 5 / Vraag 5

Find the horizontal asymptote(s) of the function / Vind die horisontale asimptoot (te) van die funksie

$$f(x) = \frac{e^x}{e^x - 7}$$

$$\lim_{x \rightarrow \infty} \frac{e^x}{e^x - 7}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 - 7/e^x}$$

$$= 1$$

$$\lim_{x \rightarrow -\infty} \frac{e^x}{e^x - 7}$$

$$= \lim_{x \rightarrow -\infty} \frac{1}{1 - 7/e^x}$$

$$= 0$$

$\Rightarrow y = 1$ and $y = 0$ are
horizontal asymptotes.

2

Question 6 / Vraag 6

6.1 Prove that $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$ by using the derivatives of $\sin x$ and $\cos x$.

Bewys dat $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$ deur gebruik te maak van die afgeleides van $\sin x$ en $\cos x$.

$$\begin{aligned}\frac{d}{dx} \cot x &= \frac{d}{dx} \frac{\cos x}{\sin x} \\&= \frac{-\sin x \cdot \sin x - \cos x \cdot \cos x}{(\sin x)^2} \\&= \frac{-1(\sin^2 x + \cos^2 x)}{\sin^2 x} \\&= -\operatorname{cosec}^2 x\end{aligned}$$

6.2 If $f(\theta) = \frac{\cot \theta}{\ln(\theta + \sin \theta)}$, find $f'(\theta)$.

As $f(\theta) = \frac{\cot \theta}{\ln(\theta + \sin \theta)}$, vind $f'(\theta)$.

$$f'(\theta) = \frac{-\operatorname{cosec}^2 \theta \cdot \ln(\theta + \sin \theta) - \cot \theta \cdot \frac{1 + \cos \theta}{\theta + \sin \theta}}{[\ln(\theta + \sin \theta)]^2}$$

Question 7 / Vraag 7

Consider / Beskou $f(x) = \frac{e^x}{x}$

7.1 Determine the critical number(s) of f . / Bepaal die kritieke getal(le) van f .

$$f'(x) = \frac{e^x \cdot x - e^x}{x^2} = 0$$

$$\Rightarrow e^x(x-1) = 0$$

$\Rightarrow x = 1$ is a critical number.

[$x = 0$ is not
in the D_f and thus
not a critical number.]

7.2 Use the first derivative test to determine the local extreme values of f on the interval $(0, \infty)$.

Gebruik die eerste afgeleide toets en bepaal die lokale ekstreemwaardes van f op the interval $(0, \infty)$.

$$\begin{array}{ccc} f' < 0 & f' = 0 & f' > 0 \\ \hline & | & \\ & \downarrow & \end{array}$$


There is a local minimum at

$$x = 1 \quad \text{namely} \quad \underline{f(1) = e}$$

Question 8 / Vraag 8

Find the absolute extreme values of f on the given interval if
Vind die absolute ekstreemwaardes van f op die gegewe interval as

$$f(x) = x - \ln x, \quad \left[\frac{1}{2}, 2\right]$$

(Hint: / Wenk: $\ln 2 \approx 0.7$)

interval method, f continuous on $\left[\frac{1}{2}, 2\right]$ - use Closed

$$f'(x) = 1 - \frac{1}{x} = 0$$

$$\Rightarrow x - 1 = 0 \Rightarrow x = 1 \text{ is a critical number.}$$

$$f\left(\frac{1}{2}\right) = \frac{1}{2} - \ln \frac{1}{2} = \frac{1}{2} + \ln 2 = 0.5 + 0.7 = 1.2$$

$$f(1) = 1$$

$$f(2) = 2 - \ln 2 = 2 - 0.7 = 1.3$$

$\Rightarrow$ Abs. max 1.3 at $x = 2$
and Abs min 1 at $x = 1$

Question 9 / Vraag 9

f is a differentiable function and $g(x) = xf(x)$.

Use the definition of a derivative to prove that $g'(x) = f(x) + xf'(x)$.

f is 'n differensieerbare funksie en $g(x) = xf(x)$.

Gebruik die definisie van 'n afgeleide om te bewys dat $g'(x) = f(x) + xf'(x)$.

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)f(x+h) - xf(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{xf(x+h) + hf(x+h) - xf(x)}{h}$$

$$= \lim_{h \rightarrow 0} x \left[\frac{f(x+h) - f(x)}{h} \right] + \lim_{h \rightarrow 0} \frac{hf(x+h)}{h}$$

$$= x \cdot f'(x) + f(x)$$